SAMPLE PAPER
B.Tech. Comp. Engg. Vth Sem Examination
AUTOMATA THEORY
Paper no CEN-502

Time : Three hours **Max. Marks: 60**

Q1 a) Describe the characteristics of an Automaton with e.g. 6 Marks

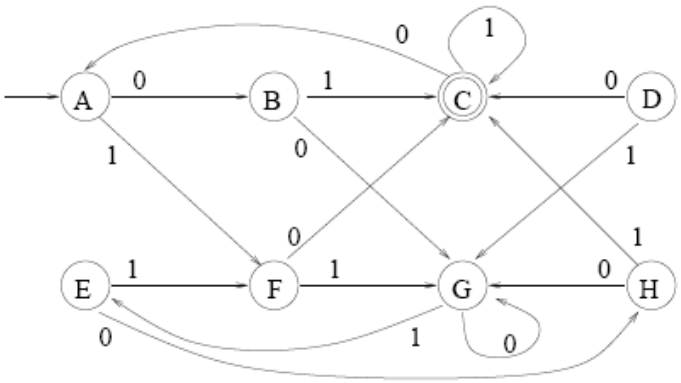
Q1 b) Construct a DFA for the followings and validate your answers: 6 Marks
i) a(aa+bb)*aabb
ii) 1((0+11)*+00)*1

OR

Q1' b) Construct Mealy and Moore machines that output '1' whenever it receives input as three consecutive '1'. 6 Marks

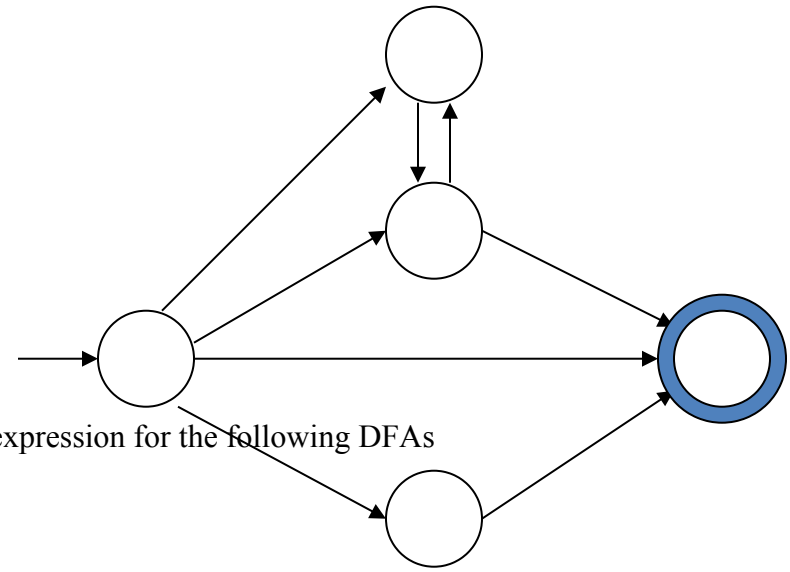
Q2 a) Comment on the following statements in terms of their validity. Give reasons for your answers. 6 Marks
i) Every subset of a regular set is regular
ii) Every finite subset of non-regular set is regular

Q2 b) Minimize the following DFA: 6 Marks

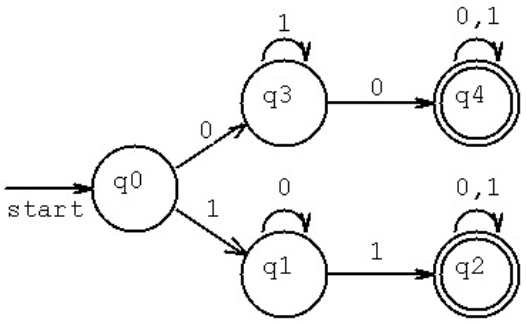


OR

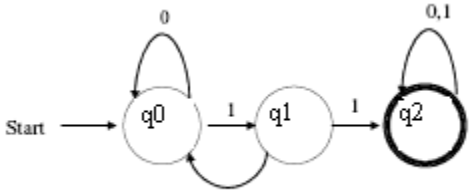
Q2' a) Convert the following NFA to DFA. 6 Marks



Q2'b) Construct the regular expression for the following DFAs 6 Marks
i)



ii)



Q3 a) Argue whether the following languages are regular or not? 6 Marks

i) L1 = {w*x*w$^+$: w, x ϵ {0, 1}$^+$}

ii) L2 = {1^n : n is even}

Q3 b) Construct a CFG for a language accepting all strings over {0, 1} which do contain substring 101. Also identify the language of the given grammar:

S → AbB

A → aA | a

B → bB | b

Q4 a) Simplify the following CFG. 6 Marks

S → ABC | aABa B → bB | CC K → ABt

A → aC | CB C → d | ε

Mention all the steps and the corresponding rules used at appropriate positions

Q4 b) Construct a PDA for the followings: 6 Marks

i) A Language accepting set of all strings over {a, b} which do not contain abab.

ii) CFG given in Ques 4a).

Validate your answers.

Q5 a) Construct a Turing Machine to identify a Language L = {a^{n}b^{n}c^n : n >=0}. 6 Marks

Also design a TM for the operation : Ц**w**Ц → Ц**w**Ц**w**Ц ,

Q5 b) Show that the f(x, y) = x^y is primitive recursive. Also explain following terms: 6 Marks

i) Recursively Languages

ii) NP Hard Problems